

Dendritic morphology as a dictionary for local credit assignment

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Abstract

Dendritic trees compartmentalize synaptic inputs, but which spatial distinctions a learning signal must preserve is unclear. We describe delivered credit as combinations of spatial patterns constrained by morphology, and compare tasks that place different demands on those patterns. Image classification benefits mainly from neuron-specific feedback, contextual conflict requires branch selection, hierarchical distractors favor ancestry at an intermediate bandwidth, and input-spectrum-matched pairwise and quartic tasks develop different spatial credit spectra and different tolerance of fixed credit profiles. Reconstructed arbors provide modeled spatial capacity beyond a common broadcast, while focal shunts act as gains on a baseline-weighted ancestry partition, with spatial selectivity depending strongly on electrotonic state. Measured visual responses show no preferential ancestry alignment, delimiting the biological evidence for a framework that links task structure to useful credit resolution without assuming that finer feedback or greater dendritic depth is always advantageous.

Introduction

Learning must link a synapse’s local activity to its contribution to a circuit outcome. Cable filtering, nonlinear integration and spatially organized inhibition make locations within a dendritic tree functionally distinct [1–7]. These differences expand computation, but when does branching also help distribute useful learning signals?

Credit assignment asks how changing a synapse would affect task error. Backpropagation (BP) answers this by calculating the loss gradient—the sensitivity of task error to each parameter [8]; related sensitivity calculations apply to recurrent and steady-state networks [9, 10]. For a point neuron, each incoming-weight gradient is the product of a locally available sensitivity, called eligibility, and a neuron-specific learning signal. Hebbian plasticity captures the association between local pre- and postsynaptic variables [11], while three-factor rules combine that eligibility with an outcome-dependent signal [12]. Here, a rule is local when a synapse uses its own eligibility and an available learning signal. This definition does not require that the signal itself be computed locally.

Biologically motivated learning theories differ in how they obtain such signals. Feedback alignment communicates errors through separate pathways [13, 14]; equilibrium, predictive and prospective methods infer target states or errors from dynamics [15–17]; eligibility propagation combines temporal traces with neuron-specific signals [18]. One number shared across neurons carries less task information than separately assigned neuronal signals, although each synapse still has its own eligibility [19–21]. Recent causal evidence for distinct instructive signals across neuronal populations motivates asking what additional distinctions are useful within a neuron’s arbor [22].

We describe this organization using a *route dictionary*: a set of spatial patterns for delivering learning signals. One pattern broadcasts throughout an arbor; several can distinguish subtrees; a complete set can represent every compartment’s error. A pattern specifies the relative signal at each location, and its coefficient sets the amount delivered on a given trial. Their linear combinations give the available fields. We use *basis* for linearly independent patterns and *dictionary* when patterns may be redundant. Morphology constrains the available patterns; the task and teaching mechanism determine which combinations are useful.

The same anatomical address can transmit a learning signal with different gains as conductance changes. Excitatory and inhibitory contacts open nonnegative conductances whose reversal potentials determine current.

Inhibition can lower input resistance and attenuate other inputs even when its own current is small [23–27]. Conductance state also changes how a local voltage change affects somatic output and task error. A change in channel amplitude rescales a delivery profile; a change in cable transport can reshape the profile itself.

Dendritic learning theories already provide mechanisms for somatic prediction, segregated basal and apical teaching, interneuron-mediated errors, burst-dependent plasticity and prospective dynamics [28–34]. Detailed cable-learning models use synaptic sensitivities and local approximations to optimize a reconstructed neuron’s input–output function [35]. Our question concerns the resolution and geometry of feedback over a whole arbor: when does a morphology-constrained dictionary preserve the distinctions a task requires, and when is a simpler delivery rule sufficient?

The conductance-tree gradient factorization and LocalCA rule were introduced in our predecessor preprint [36]. Here we examine how task structure changes the spatial distinctions that credit must preserve, and connect those distinctions to anatomical routes and conductance-dependent transport.

We test a task-to-credit progression: preserving neuron identity on image tasks, selecting the relevant branch under contextual conflict, grouping related branches at limited bandwidth, and resolving example-dependent credit for interactions across branches. Morphology supplies candidate spatial patterns, while conductance controls how those patterns are expressed. Forward representability provides a necessary starting point for interpreting learning, and measured responses test the reach of the anatomical proposal. The central question is which distinctions a useful learning signal must preserve, rather than whether finer feedback is always better.

Results

Exact gradients separate local sensitivity from downstream credit

A useful synaptic update combines local sensitivity with the effect that the change would have on task error. For a point neuron $y_u = f_u(\sum_i w_i x_i + b_u)$, the chain rule gives $\partial \mathcal{L} / \partial w_i = x_i f'_u \delta_u$, where x_i and w_i are synaptic input and weight, b_u is bias, and $\delta_u = \partial \mathcal{L} / \partial y_u$ is the neuronal error. The derivative f'_u is evaluated at the current input sum. Thus local eligibility $x_i f'_u$ and task-derived credit supply different information (Fig. 1A,C).

We replace the point neuron by a rooted tree with spatially distinct addresses, with soma $n = 0$, compartment voltage V_n , and nonnegative synaptic conductances g_i . A contact with reversal potential E_i^{rev} and nonnegative activity x_i contributes current $g_i x_i (E_i^{\text{rev}} - V_n)$. In the directed artificial trees, child outputs drive their parents and a terminal-to-soma pass computes the steady state. Total input resistance R_n^{tot} is the reciprocal of the summed leak, excitatory, inhibitory and child-coupling conductances (Methods). Differentiating the current balance yields

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\delta_{n,u}^V}_{\partial \mathcal{L} / \partial V_n}. \quad (1)$$

Both excitatory (E) and inhibitory (I) contacts obey this factorization (Fig. 1A). Eligibility tells how conductance changes local voltage; compartment credit tells how that voltage changes task loss.

Here voltage error is a loss derivative, not a voltage difference. In a directed tree, each compartment has one path to the soma. Its exact voltage error is

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad (2)$$

Here $f'_i(V_i)$ is the child compartment’s output derivative, $g_{i \rightarrow k}^{\text{den}}$ is its coupling conductance to parent k , and $\delta_{0,u}^V = f'_0(V_0) \delta_u$ is the somatic voltage error. Thus the path multiplier $\tilde{\alpha}_n$ converts one somatic error into a spatial field. The conductance-only component of this product is attenuated by additional conductance on a path’s ancestors; branch nonlinearities and forward voltages can change as well (Supplementary Section S1).

Reciprocal cables instead require a joint adjoint calculation of how local currents affect task loss (Methods). These mathematical sensitivities do not establish a biological error-transmission mechanism.

Our local credit assignment (LocalCA) rule retains the local sensitivity but supplies an approximate learning signal to each compartment. A hat denotes this delivered field:

$$\widehat{\mathbf{V}}_{g_i} \mathcal{L} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \widehat{\boldsymbol{\delta}}_{n,u}^V, \quad (3)$$

$$\widehat{\boldsymbol{\delta}}_u^V = A_u \mathbf{c}_u. \quad (4)$$

Columns of $A_u \in \mathbb{R}^{N_u \times K}$ are fixed spatial delivery profiles over N_u routed nonsomatic compartments; $\mathbf{c}_u$ contains their example-dependent coefficients. Broadcast, subtree and compartment profiles give increasingly detailed spatial addresses (Fig. 1B). This linear delivery model does not follow automatically from the forward cable equations and does not specify how a circuit estimates the coefficients.

At $K = 1$, the all-ones column $A_u = \mathbf{1}$ broadcasts one coefficient through the arbor; at $K = N_u$, the identity matrix $A_u = I$ can reproduce the full nonsomatic field. Nested subtrees provide intermediate dictionaries. Profile count, rank (the number of independent profiles) and coefficient information are distinct resources. Exact path transport combines one somatic scalar with state-dependent local derivatives; it does not require an independent external error for every compartment.

Capture is the fraction of squared learning-signal magnitude that a dictionary can reproduce. For a nonzero field $\boldsymbol{\delta}$,

$$\mathcal{C}(A; \boldsymbol{\delta}) = \frac{\|P_A \boldsymbol{\delta}\|^2}{\|\boldsymbol{\delta}\|^2}, \quad P_A = AA^\dagger.$$

Here P_A gives the best least-squares reconstruction from the profiles; $A^\dagger$ is the Moore–Penrose pseudoinverse and $\|\cdot\|$ the Euclidean norm. Thus capture is a fraction of squared field magnitude, not a fraction of correct decisions. The mean of these per-field ratios weights fields equally. In contrast, $\text{tr}(P_A C) / \text{tr}(C)$, with $C = \mathbb{E}[\boldsymbol{\delta} \boldsymbol{\delta}^T]$, is the fraction of total ensemble energy captured, weighting fields by their energy. Here C is the uncentered second-moment matrix and tr denotes its trace, the sum of diagonal entries. The two capture measures need not agree. Projection chooses the best coefficients in the specified metric; an implemented learning rule need not supply them.

The constructions below share this expansion principle but differ in their coordinates, coefficient sources and state dependence; Supplementary Table S14 compares them. Dictionary capacity answers what fields can be delivered. Learning experiments determine which of those distinctions improve task performance.

One signal per neuron captures most of the image-task learning benefit

We compared three levels of feedback: one signal shared across neurons, one signal per neuron broadcast through its own arbor, and the exact spatial field (Fig. 1B,C). The decoder supplied exact somatic errors in all three conditions. The strict scalar s_i^{sc} used their population mean, with a largest-magnitude fallback when the mean nearly cancelled (Methods). Forward networks were otherwise identical.

The primary comparison used flattened MNIST and 128 directed [3, 3] trees, each with three proximal and nine terminal compartments. Shunting and topology-matched raw-additive architectures shared the branching graph, contact masks, nonlinearities, decoder and training protocol, with architecture-specific initialization. The additive control retained signed current summation but removed reversal-potential driving forces and the total-conductance denominator. It therefore could not shunt.

Neuron-specific feedback improved accuracy over the strict scalar by 11.2 percentage points in shunting trees and 7.8 points in raw-additive trees. Exact path transport then changed accuracy by less than 0.2 points in either architecture (15 paired seeds; Fig. 1D,E). Most of the scalar-to-exact improvement therefore came from preserving distinct neuronal coordinates. Fashion-MNIST also favored neuron-specific feedback over the matched-width scalar-fallback control. On flattened CIFAR-10, neuron-specific feedback outperformed the strict scalar, but exact path transport reduced accuracy relative to neuron-specific feedback by 0.86 points (95% paired interval, 0.55–1.17 points; Fig. 1F). Exact transport remained equivalent to matched backpropagation within the predefined margin (Supplementary Fig. S4).

114 The historical matched-width fallback retained neuron identity for some near-somatic parameters, improving
 115 training. The main MNIST comparison therefore uses the strict scalar (Methods); local eligibilities remain synapse
 116 specific in either case.

117 At common checkpoints, exact factorized transport reproduced automatic differentiation; gradient-cosine
 118 diagnostics locate the approximation in the delivered field (Supplementary Fig. S45C).

119 The exact field nevertheless contained substantial spatial structure. Shunting attenuated error magnitude distally,
 120 whereas the raw-additive field amplified it (Supplementary Fig. S45D). After subtracting the mean across paths at
 121 each depth, path-specific residuals accounted for about 18% of error energy at the middle stage in both architectures,
 122 rising to 32% distally in additive trees and 53% in shunting trees. The distal shunting fraction was larger in all fifteen
 123 paired seeds (Supplementary Fig. S45E), yet resolving it brought little additional accuracy on MNIST.

124 We also separated neuron identity from *ownership*, the delivery of each coordinate to its corresponding arbor
 125 (Supplementary Fig. S45F). Fixed permutations among arbors preserved bandwidth and coordinate values but
 126 reduced accuracy by 0.57 and 0.51 points in two-stage shunting and additive trees, and by 0.38 and 0.39 points at
 127 four stages. All ten paired seeds favored correct ownership in each condition (Supplementary Fig. S45G). These
 128 separate cohorts test recipient identity rather than within-arbor resolution.

129 The result survived replacing exact between-neuron feedback with a fixed random soma-level matrix. Under
 130 direct feedback alignment, neuron-specific feedback improved over a scalar by 10.8 points in additive trees and 11.0
 131 points in shunting trees, while exact path transport added at most 0.09 points. At the neuron and path rungs, random
 132 feedback trailed the exact decoder map by no more than about 0.2 points (15 paired seeds; Supplementary Fig. S30).
 133 The within-arbor ordering therefore did not require an exact between-neuron teaching pathway.

134 These image benchmarks establish that a spatially structured error field need not translate into a benefit from finer
 135 feedback. Supplementary Figs. S4 and S8 retain further depth and task-generator controls. The next experiments
 136 introduce task structure that makes particular spatial distinctions useful. A one-step smoothness bound explains how
 137 restricting noisy updates can improve expected progress, but its initialization utility lost to a fixed maximum-budget
 138 tree in all 20 seed averages of a 12,800-fit prospective selection test, in both selection arms. The bound and noise
 139 screens therefore describe local update quality rather than identify an optimal morphology (Methods; Supplementary
 140 Sections S3–S4).

141 **Branch-selective information prevents updates from inactive distractors**

142 A learning signal can be harmful if it reaches a branch that received an input but did not contribute to the output.
 143 To test branch selection directly, we used one logistic output unit with $B \in \{2, 4, 8\}$ parallel branch weight vectors
 144 and no internal forward compartments or conductance nonlinearity (Fig. 2A). A context selected the branch that
 145 determined the output, while every branch received an image and formed an ungated local eligibility. This separates
 146 the information needed to select a branch from any specifically dendritic implementation.

147 Inputs were average-pooled, standardized Fashion-MNIST images from classes 0 and 6. The selected image
 148 determined the binary target. At conflict dose $\chi = 0$, all branches received that image; increasing χ independently
 149 replaced nonselected inputs with opposite-class images. For example i , let b_i^* be the selected branch, $\mathbf{x}_{i,b}$ its input
 150 vector and $y_i \in \{0, 1\}$ the target. The branch weights $\mathbf{w}_b$ give the pre-probability score (logit) and output error

$$z_i = \mathbf{w}_{b_i^*}^\top \mathbf{x}_{i,b_i^*}, \quad \delta_i = \sigma(z_i) - y_i. \quad (5)$$

151 Here $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function. The exact derivative of every unselected branch is zero, irrespective
 152 of its input. Increasing conflict therefore does not create opposing nonzero exact branch gradients. It makes a rule
 153 that ignores the selector increasingly sensitive to irrelevant, opposite-class eligibilities.

154 For N examples, correctly selected and neuron-shared gradient estimates were

$$\mathbf{d}_b^{\text{selected}} = N^{-1} \sum_i \delta_i \mathbf{1}[b_i^* = b] \mathbf{x}_{i,b}, \quad \mathbf{d}_b^{\text{shared}} = N^{-1} \sum_i \delta_i B^{-1} \mathbf{x}_{i,b}. \quad (6)$$

155 The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. The factor $1/B$ matches scale but supplies
 156 no branch information. The selector can be delivered in feedback or supplied by a branch-local eligibility gate; the

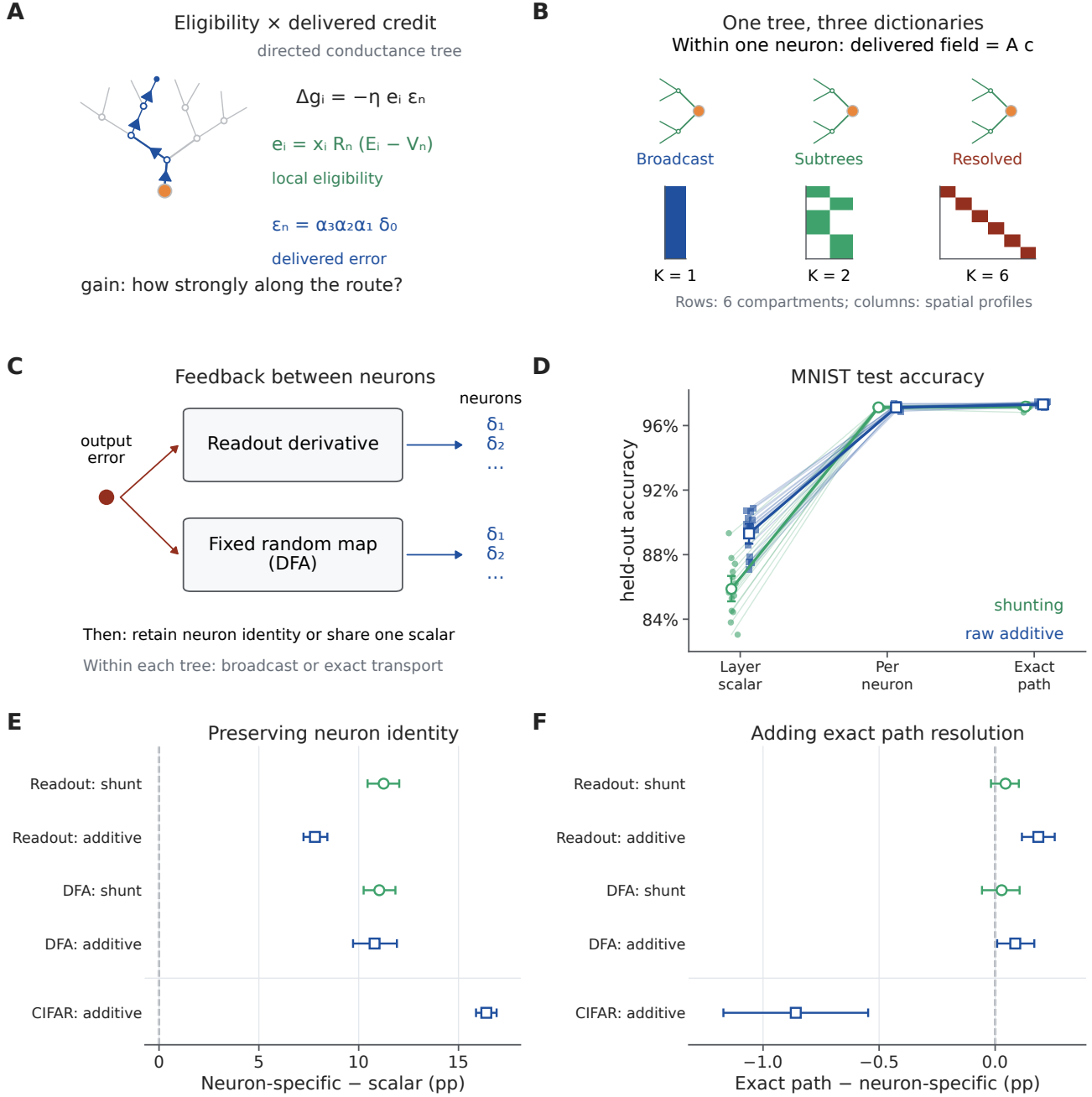


Figure 1: Teaching coordinates and spatial resolution make distinct contributions to image learning. **A**, Conductance updates combine local eligibility with delivered credit. Symbols denote learning rate η , presynaptic activity x_i , input resistance R_n , driving force $E_i - V_n$ and delivered voltage error ϵ_n . Directed exact transport multiplies somatic error δ_0 by path derivatives α ; reciprocal cables require an adjoint. **B**, Six nonsomatic compartments support one broadcast, two subtree or six compartment profiles. Matrix A places the K profiles and c supplies example-dependent coefficients. Exact transport can resolve the field using one external somatic error and local state. **C**, Readout derivatives or fixed random direct feedback alignment (DFA) supply neuronal coordinates before within-arbor delivery. **D**, MNIST test accuracy in shunting and raw-additive trees; thin lines pair seeds. **E,F**, Paired accuracy gains from preserving neuron identity and adding exact path resolution, respectively. Symbols and bars are means and 95% intervals; colors identify architectures. MNIST has fifteen paired seeds per architecture and feedback source; flattened CIFAR-10 has twenty. Percentage-point (pp) axes differ between panels. The small additive-MNIST increment and negative CIFAR-10 increment delimit the predominance of neuron identity.

157 resulting update is identical (Fig. 2B). Consequently, the experiment tests access to branch-selective information, not
158 a requirement for several independently computed external error values.

159 A simplified calculation asks when the selected branch’s evidence is outweighed by conflicting evidence
160 sent to other branches. With centered, symmetric class evidence, the shared-mode prediction crosses zero at
161 $\chi_c = B/[2(B - 1)]$: 1, 2/3 and 4/7 for two, four and eight branches (Fig. 2C; Methods). More branches therefore
162 predict an earlier loss of compatibility. This average calculation is distinct from the conflict dose at which a trained
163 model reaches chance accuracy.

164 At zero conflict, selected and shared rules achieved similar accuracy. Their difference increased with conflict,
165 reaching 35–58 percentage points at full conflict across the tested branch counts (20 paired seeds; Fig. 2D). In
166 every seed, the advantage of correct routing increased with conflict. A cyclic reassignment sent every signal to the
167 wrong branch at matched rank and sparsity and failed at high conflict. Analytic backpropagation and an explicitly
168 gated grouped-point model matched correct selection exactly, as required by their algebraic equivalence (Fig. 2E;
169 Supplementary Fig. S29).

170 The shared rule reached chance at lower conflict doses as branch count increased, consistent with the predicted
171 shift (Fig. 2F). Five two-branch seeds did not cross in the sampled range. Mean-curve, individual-seed and theoretical
172 crossings are different quantities. Checkpoint diagnostics also showed that gradient alignment changed during
173 learning (Methods; Supplementary Fig. S29D–F). The causal result is that sending an error to branches that received
174 inputs but did not determine the output can harm learning. A dendritic branch is one possible site for this selection.

175 **Ancestry provides a modest advantage at an intermediate feedback budget**

176 Branch selection compares fully shared with fully resolved feedback. We next asked whether nested subtree routes
177 offer a useful intermediate dictionary. The task contained eight independent, eight-dimensional terminal weight
178 blocks at the leaves of a balanced binary feedback tree. Only the context-selected block determined the logit. Internal
179 tree nodes defined feedback supports and did not perform serial forward computation (Fig. 3A).

180 The selected stream carried the class signal. Distractors carried opposing signals that strengthened with distance
181 through the feedback hierarchy, deliberately favoring task-aligned grouping. At each budget $K \in \{1, 2, 4, 8\}$, the
182 routing field had K independent patterns and unit-length rows. Ancestry routes grouped the terminals into the
183 whole tree, halves, sibling pairs or individual terminals. Permuting task streams across leaves changed alignment
184 without changing the streams or forward parameter budget. Here bandwidth counts independent feedback patterns,
185 not physical stages or independent scalar losses; other single-pattern controls need not be uniform. The complete
186 generator and update are given in Methods.

187 The generator predicts where an intermediate ancestry benefit can occur. The selected stream has class coefficient
188 one; its sibling, same-half nonsiblings and other-half streams have coefficients -0.15 , -0.45 and -0.75 . The
189 unnormalized sum of the class signal within the selected ancestry group gives -3.05 , -0.05 , 0.85 and 1 at
190 $K = 1, 2, 4, 8$, respectively. Thus coarse pooling reverses or nearly cancels the signal, sibling-pair routing first
191 preserves a positive intermediate-group signal, and full resolution removes the grouping distinction. The implemented
192 unit-row normalization divides these sums by the square root of group size, preserving their signs. This is an analytic
193 consequence of the fixed task design; its location is not an independently discovered optimum.

194 Controls isolated different resources. Cyclic derangement preserved ancestry supports, rank and sparsity but
195 assigned signals to the wrong groups. Depth-interleaved and random-sparse fields preserved rank and sparsity
196 while changing support geometry. Dense random fields preserved rank and row norm while relaxing sparsity. A
197 low-rank reference derived by principal component analysis (PCA) used training information but did not bound
198 trained accuracy. The main best-control summary was the per-seed maximum over deranged, depth-interleaved,
199 random-sparse and dense random fields, computed after training. It was a conservative comparator, not a route
200 chosen by the model (Fig. 3C,D).

201 Increasing bandwidth improved correct routing, while derangement caused large losses (Fig. 3C). The ancestry-
202 specific effect was much smaller. Ancestry performed worse than the best matched control at low budgets, exceeded
203 it by 1.27 percentage points at $K = 4$, and tied at full resolution. The paired $K = 4$ interval was 0.59–1.95 points,
204 with 15/20 positive differences; a paired Wilcoxon signed-rank test gave Holm-adjusted $P = 0.0101$ across four

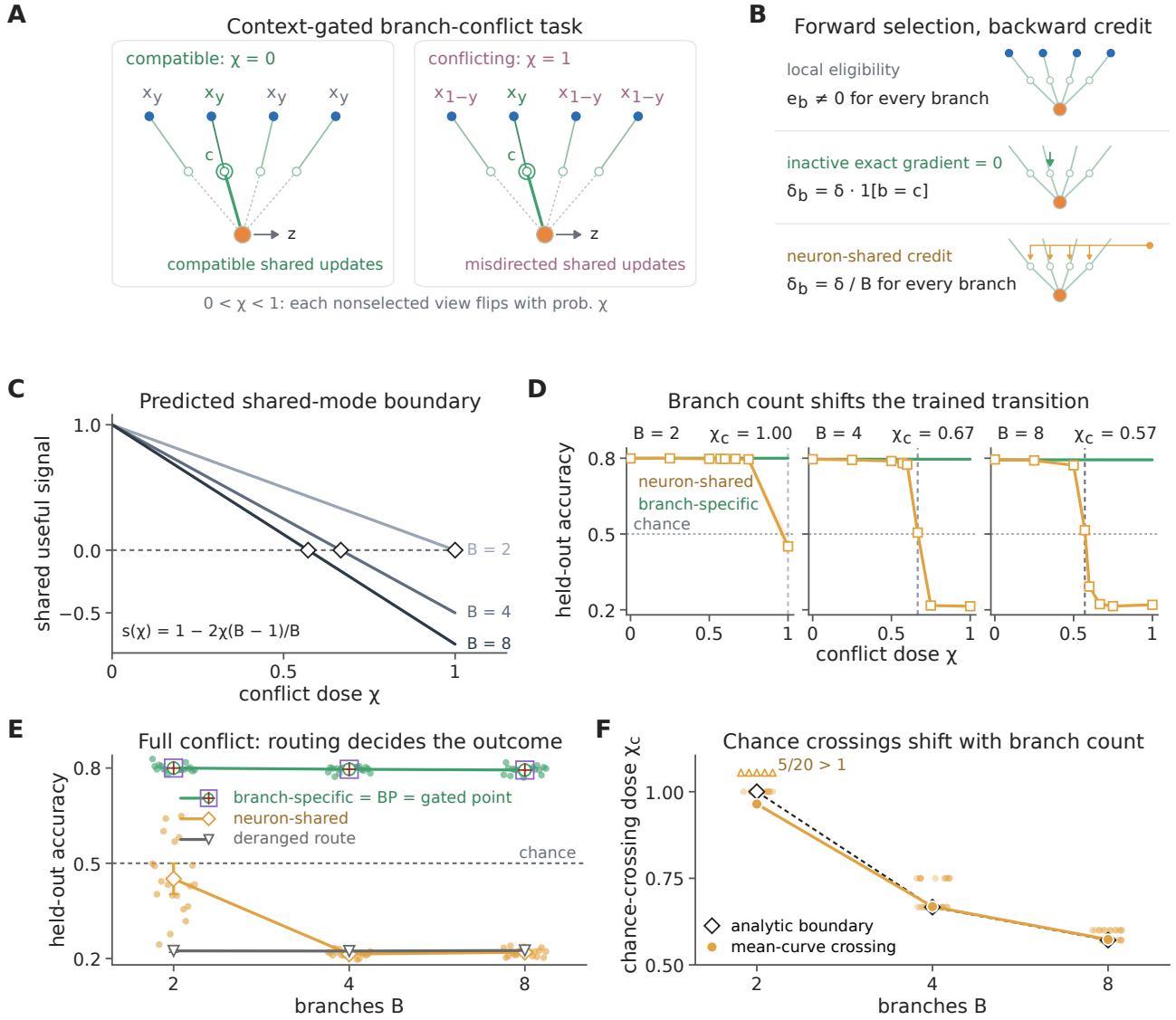


Figure 2: Branch selection prevents learning from inactive distractors. One logistic output combines B parallel branch weight vectors; there are no serial or conductance nonlinearities. **A**, Context-gated Fashion-MNIST task. Context c selects the branch determining logit z ; every branch receives an image and forms an ungated eligibility. Conflict dose χ independently replaces each nonselected image by an opposite-class image. Solid and dashed proximal paths are selected and gated-off forward routes; the double ring marks context. **B**, Selected feedback: $\delta_i \mathbf{1}[b = b_i^*]$, shared feedback: δ_i / B . Here δ_i is the output error, b_i^* the selected branch and $\mathbf{1}$ the indicator; unselected exact derivatives are zero. A local eligibility gate gives the same update. **C**, Centered symmetric-evidence prediction $s(\chi) = 1 - 2\chi(B - 1)/B$: shared-mode compatibility vanishes at $\chi_c = B/[2(B - 1)]$. This mean-field boundary is distinct from a trained chance crossing. **D**, Held-out accuracy over conflict dose. **E**, At full conflict, correct selection, backpropagation (BP) and the algebraically equivalent gated-point calculation coincide; cyclic derangement sends signals to the wrong branch at matched rank and sparsity. Selection-by-conflict interactions are positive in all 20 seeds for each B . **F**, Analytic threshold (diamonds), first interpolated crossing of mean accuracy (orange), and each seed's first sampled at-or-below-chance dose (faint dots). Open overflow triangles retain five $B = 2$ seeds that do not cross within the sweep. Curves and bars summarize 20 paired seeds with 95% seed-bootstrap intervals. Branch-count comparisons do not hold capacity fixed.

budgets. Individual control contrasts and paired seed differences distinguish this effect from the much larger correct-versus-deranged contrast (Fig. 3E,F).

Degree- and depth-preserving rewiring removed the task-aligned advantage at intermediate budgets but did not affect fully shared or fully resolved ancestry routes (Fig. 3D). Flat, grouped-point and gated-point representations receiving the same field behaved identically by construction. The experiment therefore supports a conditional advantage of ancestry-shaped feedback under limited bandwidth, without attributing the effect to dendritic material or treating equivalent representations as independent confirmations.

The enlarged four-route effect and individual-control comparisons show the modest size of the ancestry benefit (Fig. 3E,F). The trained budget profile agrees with the coefficient-derived prediction, while overall accuracy still favors full bandwidth. The nearly eight-dimensional initial credit spectrum follows from balanced context representation and is not an independent morphology predictor.

Learning the route coefficients was another limitation. In twenty fresh seeds, an estimator trained on separate noisy context cues improved accuracy over a frozen profile by 7.16 points, but remained 54.47 points below oracle context coefficients. Exploratory selection of its most probable route narrowed the gap, whereas delaying cues removed the benefit (Supplementary Fig. S32; Methods). These tests expose requirements for selective, timely delivery; they do not derive the context signal from local eligibility.

Cross-branch interactions change the spatial credit that learning uses

Context and ancestry determine which branches should receive a signal. Interactions between branches can also change the signal’s sign and magnitude from example to example. We tested this distinction on tasks with identical input-sensitivity spectra, after ensuring that the same tree could represent both targets. This separates a limitation of credit delivery from a limitation of the forward computation.

Eight independent inputs x_i took values -1 or $+1$ with equal probability. Each branch combined child outputs as $u = a + bu_L + cu_R + du_Lu_R$, with four trainable coefficients. Such a unit is multi-affine: it is affine in either child while the other is fixed. Pairwise targets summed four disjoint products, $f_M = \frac{1}{2} \sum_{(i,j) \in M} s_{ij} x_i x_j$; quartic targets summed products over two complementary quartets, $f_A = \frac{1}{2} (s_A \prod_{i \in A} x_i + s_{\bar{A}} \prod_{i \notin A} x_i)$. Signs s were independently sampled. Within each seed, both targets shared a compatible balanced tree, input assignment, initial weights and example streams (Fig. 4A). Both have input-sensitivity second moment $\mathbb{E}[\nabla_{x,f} \nabla_{x,f}^T] = I_8/4$: the strengths of independent input-sensitivity directions are identical.

Compatibility has an exact criterion in this model. At each subtree boundary, arrange the target’s interaction coefficients in a matrix indexed by products of inputs inside and outside that subtree. After removing the constant inside row, this matrix must have rank at most one, because only one scalar leaves the branch. This condition is also sufficient when it holds at every boundary; discarded singular values bound representation error (Methods, Eq. 13; Supplementary Section S4). It is a prerequisite for the credit comparison, not a theorem about the learning rule. Boolean examples and the complete structure census are retained in Supplementary Figs. S36–S37 and S43–S44.

We compared exact path credit with unit broadcast and an initially calibrated spatial profile. The calibrated rule averaged each student’s path derivatives over 256 unlabeled initialization examples and kept that six-site profile fixed during learning. It could therefore preserve initial signs and relative gains without access to the target. All rules used exact local eligibilities and the same scalar output error. Three development seeds selected learning rates with equal tuning budgets; twenty fresh paired seeds then tested the frozen choices. A separate nested-interaction target used its own compatible tree.

After 1,024 Adam steps, exact and calibrated credit achieved mean held-out normalized mean squared error (NMSE) of 0.0317 and 0.0236 on pairwise targets, compared with 0.1516 and 0.9813 on quartic targets (Fig. 4B,C). NMSE divides squared prediction error by the clean target variance. The predefined task-by-credit contrast was the calibrated-minus-exact gap for quartic targets minus the same gap for pairwise targets: 0.838 NMSE (95% paired seed-bootstrap interval, 0.719–0.931), positive in all twenty seeds. A common learning rate gave 0.843 (0.727–0.935; Fig. 4E). On nested targets, exact and calibrated NMSE were 0.0230 and 0.7690 (Fig. 4D). These are fixed-budget comparisons; some exact quartic fits retained appreciable error. Equal additive label noise produced different NMSE floors because pairwise and quartic targets have variances one and one-half, respectively.

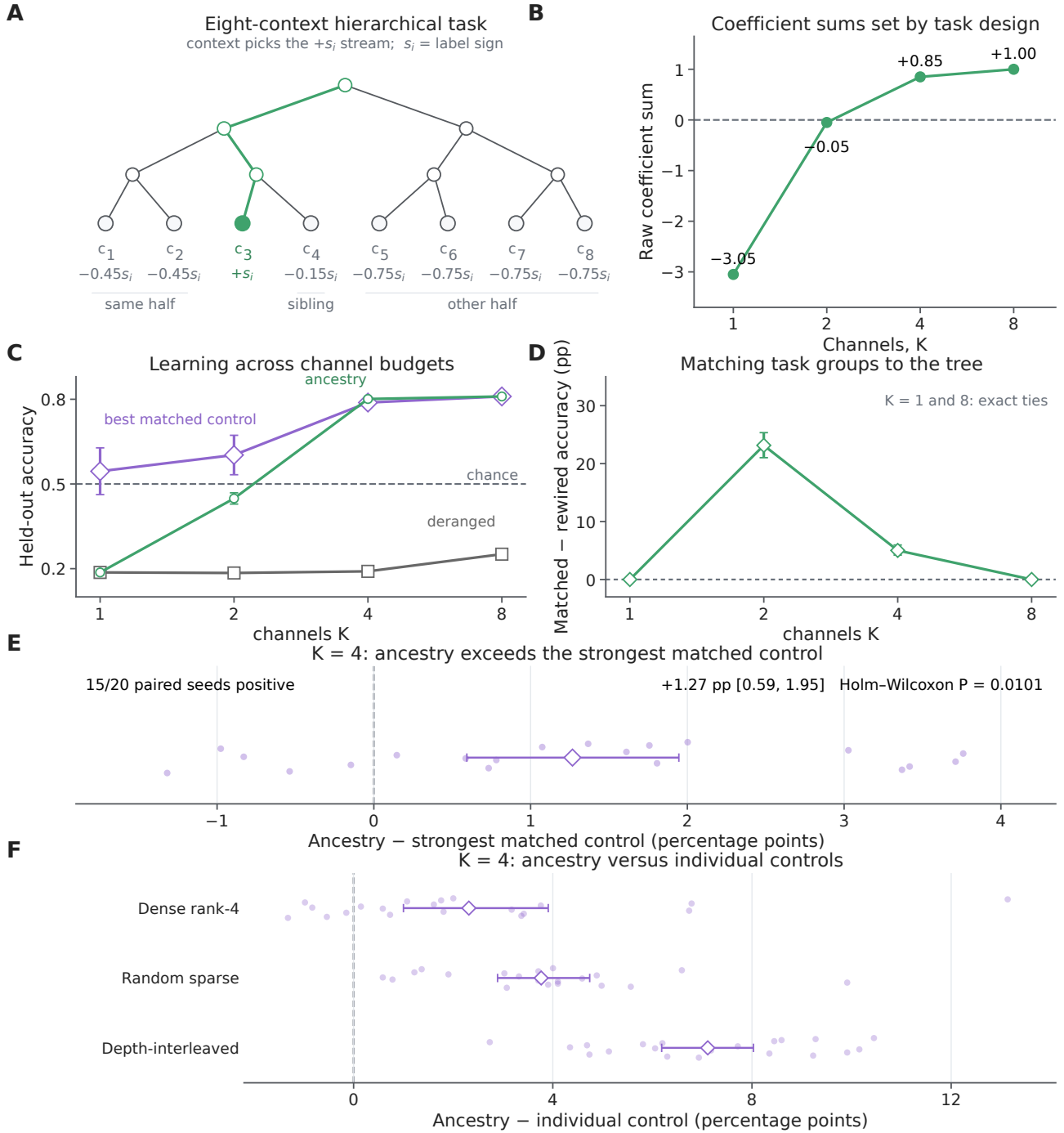


Figure 3: Hierarchical distractors favor ancestry at intermediate bandwidth. **A**, Eight streams carry label sign s_i along noisy teacher features. The selected stream has coefficient one; sibling, same-half and opposite-half distractors have coefficients -0.15 , -0.45 , -0.75 . **B**, Coefficient sums in the cued ancestry group at $K = 1, 2, 4, 8$. Implemented feedback divides these sums by square root of group size, preserving their signs. The four-route sign change follows from the design; it is not an independently discovered accuracy optimum. **C**, Held-out accuracy for correct ancestry, derangement and the per-seed maximum over four matched controls. This maximum is calculated after training, not learned by a selector. Dashed line: chance. **D**, Task-matched-minus-rewired accuracy at equal degree and depth. **E**, The four-route ancestry-minus-best-control effect, 1.27 pp (95% interval, 0.59–1.95), with 15/20 positive seeds and Holm-adjusted two-sided Wilcoxon $P = 0.0101$ across four budgets. **F**, Individual dense, sparse and depth-interleaved contrasts on a separate scale. The large derangement effect is shown in **C**. Symbols and bars are means and 95% seed-bootstrap intervals; small points in **E,F** are twenty paired seeds. Endpoint ties follow from coincident operators.

253 The learned credit fields explain a distinction that the input spectrum misses. Let $\mathbf{q}(x)$ contain the six nonsomatic
254 derivatives of root output with respect to branch output. Exact branch credit is scalar output error times $\mathbf{q}$. We
255 measured how much of its ensemble energy the best single fixed spatial profile could reproduce, allowing an
256 oracle amplitude for each example. At exact-trained checkpoints, that profile captured 99.7% of pairwise path-field
257 energy, but only 46.1% for quartic and 44.8% for nested targets (Fig. 4F). Eligibility-weighted update reconstruction
258 and site-normalized spectra showed the same separation; a replay of 600 earlier fits replicated it (Supplementary
259 Section S4). Uniform-profile capture did not separate the families: a nearly one-dimensional field can have a signed,
260 nonuniform direction.

261 Thus identical input spectra can accompany different learned credit spectra on the same tree, and fixed-profile
262 learning fails where the trained field varies in more spatial directions. At identical weights and inputs, however, path
263 derivatives are target-independent. The spectrum is a property of the learner’s state and coordinates, not a task-only
264 initialization diagnostic. These observations support a role for example-dependent spatial credit without establishing
265 that all six independent channels, or exact gradients, are necessary.

266 The effect has a useful boundary. On a compatible positive-conductance teacher, every tested fixed-profile
267 rule learned accurately with Adam; exact credit added no practical accuracy benefit. Boolean controls likewise
268 distinguished cases: broadcast learned aligned XOR-of-AND, whereas parity showed a large exact-credit advantage
269 under the tested optimizer (Supplementary Figs. S42–S44). Interaction structure therefore identifies comparisons
270 worth testing, rather than a universal monotonic law of credit resolution. Estimating interactions from noisy examples
271 also yielded useful trees; the estimator and its learning pipeline remain supporting structure-design results in
272 Supplementary Figs. S41–S42.

273 **Serial forward computation and feedback interact with the optimization regime**

274 The interaction comparison holds a compatible forward tree fixed while changing credit. We next varied serial
275 forward structure itself. We next asked whether successive compartments can remove unwanted multiplicative
276 changes in input strength (Supplementary Fig. S31). Physical depth D_p counts forward stages; feedback resolution
277 counts distinctions in the backward field. Neither is interchangeable with the number of latent factors in the data.

278 Each network contained 64 somata and eight nonsomatic compartments per soma. D1 [8], D2 [2, 3], D3 [2, 1, 2]
279 and D4 [1, 1, 2, 2] redistributed those compartments across one to four serial stages while preserving contacts and
280 trainable resources. Branching vectors are ordered from soma to tips. A literal grouped-point control sent each
281 of the eight modules directly to the soma, removing serial composition. A raw-additive tree retained the directed
282 graph while removing divisive conductance normalization. A dense multilayer perceptron (MLP) was approximately
283 parameter matched and served as a flexible comparison, not as a contact-matched morphology.

284 Distal excitatory inputs carried class information multiplied by nuisance gains. Separate inhibitory blocks sensed
285 those gains at alignment α : zero preserved the sensor distribution but removed its trial-specific information, and one
286 reported the factor exactly. With inhibitory reversal zero, these forward sensor inputs entered the denominator of the
287 conductance equation and could support successive divisions [24, 37]. The generator floored noisy input rates before
288 multiplying them by gains; full equations, floors and sensor controls are specified in Methods.

289 Three task families separated hierarchical support from generic factor cancellation. Nested factors occupied four,
290 two and one fine-to-global groups. Flat factors used three independent fields on the same eight-group partition, each
291 with its own sensor. Their products were commutative; serial order came from placement of sensors in the model.
292 In the local-ratio task, excitation and inhibition share each module’s cumulative gain, permitting local cancellation
293 (Supplementary Section S5).

294 Under backpropagation, aligned D3 models exceeded aligned D1 models by 30.9 percentage points. Independent
295 or trial-shuffled sensors and reversed tier placement removed the depth benefit. Raw-additive trees worsened with
296 depth, and the resource-identical parallel grouped-point model did not reproduce the serial computation. More
297 flexible point networks exceeded the serial model. This is therefore a task-matched computational bias under
298 constrained architecture, not a general expressivity advantage of dendrites.

299 The feedback result depended strongly on optimization. Exact transport exceeded shared-soma transport at D3
300 by 10.96 points within the LocalCA regime, but full BP exceeded soma broadcast by only 0.64 points under the

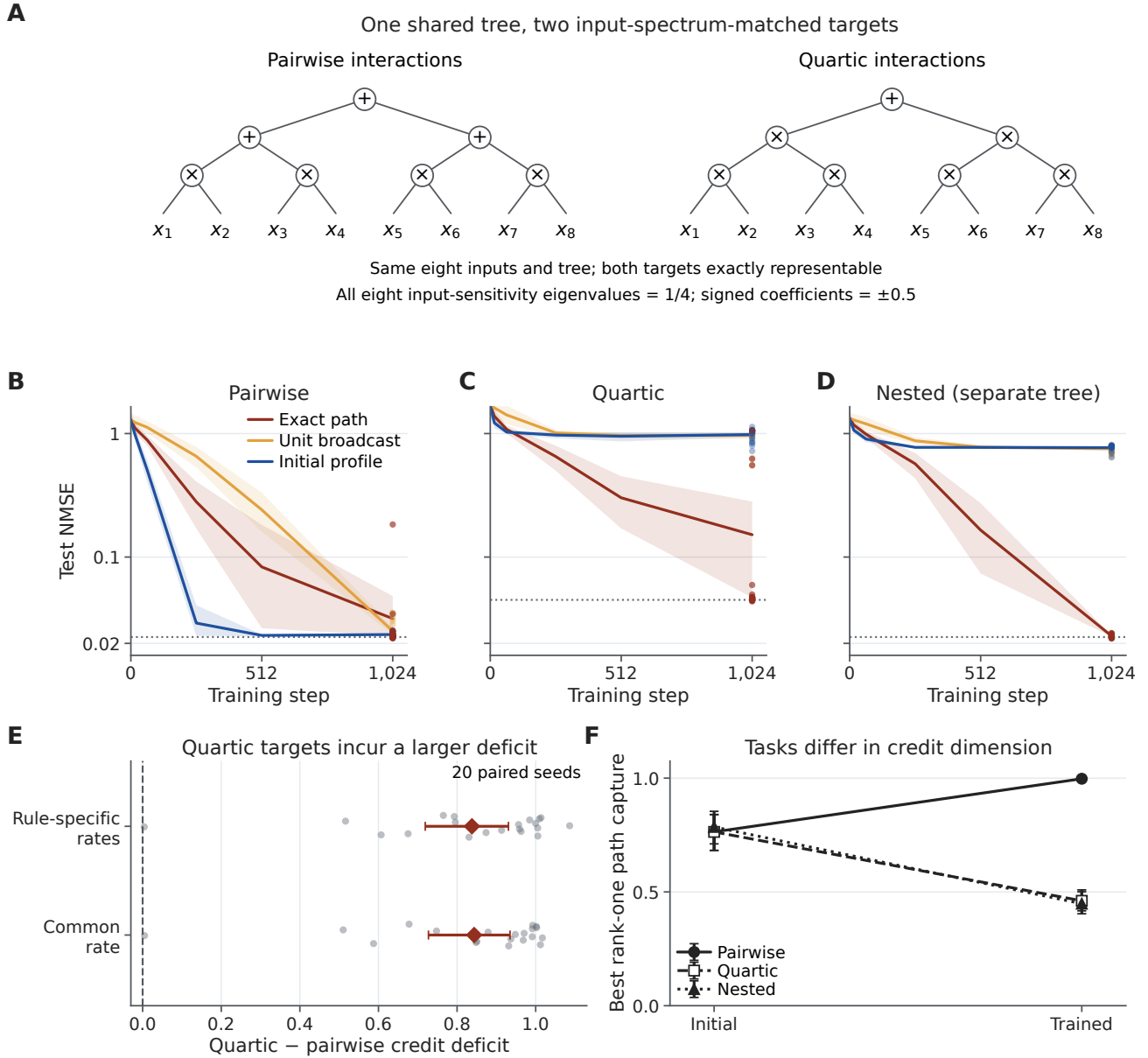


Figure 4: **Input-spectrum-matched tasks differ in learned credit geometry and tolerance of fixed spatial credit.** **A**, Pairwise and quartic targets share eight inputs, a balanced tree, initialization and example streams. Drawings show interaction supports; coefficients have random signs and magnitude 0.5, with input labels permuted together within each seed. Both targets are exactly representable and have input-sensitivity second moment $I_8/4$. **B–D**, Held-out NMSE during Adam training with exact path credit, unit broadcast and an initially calibrated profile. The latter averages the initial path derivatives on 256 unlabeled examples and remains fixed. The somatic derivative is one for all rules. Nested targets use a separate compatible tree. Rates were selected using three development seeds per rule. Lines and bands show means and descriptive 95% whole-seed bootstrap intervals; endpoint dots retain all twenty fresh seeds. Dotted lines indicate noise-only NMSE: 0.0225 for pairwise and nested, 0.045 for quartic targets. **E**, Quartic-minus-pairwise difference in the calibrated-minus-exact NMSE gap, at rule-specific rates and a common rate of 0.003. **F**, Best rank-one capture of six-site path-field energy before and after exact-credit training. The spatial direction and example-specific amplitudes are oracle fits, not an implemented encoder. Points and intervals in **E,F** summarize twenty paired seeds. All fits stop at 1,024 steps.

standard BP optimizer. Optimizer-matched controls explain why the same gradient comparison produced different endpoints (Methods; Supplementary Figs. S28 and S31). The eleven-point effect belongs to a particular optimization regime; exact-path LocalCA computes the exact selected gradients.

Depth benefits also saturated. The two-factor task favored D2, whereas the four-factor task peaked at D3 rather than D4. Serial models benefited from strongly aligned sensors in both nested and flat factor tasks, more strongly in the nested case. When ratios were locally available, BP reached ceiling in both serial and grouped models, and serial LocalCA could perform worse. Thus forward composition, sensor placement and optimization help determine the utility of a feedback dictionary.

Reconstructed arbors provide spatial routes beyond a shared broadcast

The artificial tasks show what particular credit patterns can accomplish. We next asked whether reconstructed dendritic geometry supplies useful candidate patterns. MICrONS combines electron-microscopy anatomy and connectivity with co-registered visual responses for a subset of cells [38]. Skeletons, soma positions and mapped contact locations and areas were observed; conductances, voltages and gradient fields were modeled. These analyses therefore test the capacity of measured anatomy to support spatial credit, rather than measure a teaching signal.

Each inhibitory-bearing segment supplied a candidate route over its input-bearing descendants (Fig. 5A). For a site-by-route matrix A_K , diagonal excitatory-area weights W and weighted least-squares projector $P_{A_K, W}$, capture of a field $\mathbf{t}$ was

$$\mathcal{C}(A_K; \mathbf{t}) = \frac{\|P_{A_K, W} \mathbf{t}\|_W^2}{\|\mathbf{t}\|_W^2}, \quad \|\mathbf{v}\|_W^2 = \mathbf{v}^T W \mathbf{v}. \quad (7)$$

For an ensemble of perturbations, numerator and denominator sum over field columns. A common-constrained SVD dictionary provided a ceiling at the same column budget. Controls used random anatomical routes, cable-distance bins, site-shuffled routes preserving nonzero counts, and surrogate trees preserving degree and depth while changing ancestry.

We first examined fields generated independently of the selected dictionary: modeled gradient changes caused by small focal shunts throughout a reciprocal cable. In the original eight-cell analysis, subtree routes lacked a constant component, whereas depth bins could reproduce an all-site broadcast. That difference mattered. A single broadcast captured 31.4% of total field energy, close to the 32.6% rank-one ceiling, but most field energy remained outside it. The original comparison therefore mixed the value of a common signal with that of spatial distinctions.

We corrected this resource mismatch by giving every family one budgeted broadcast column and $K - 1$ spatial profiles. We measured both total capture and capture of the field remaining after the common component was removed. The protocol was fixed before extending the analysis to 47 disjoint cells from the same mouse and an independently reconstructed second mouse. Each comparison retained actual matrix rank and nonzero coefficient count, because overlapping routes can spend the same nominal budget on different effective resources (Fig. 5).

At eight columns in the disjoint 47-cell cohort, broadcast-plus-subtree routes captured 0.619 of total energy, compared with 0.511 for depth bins and 0.436 for the site-shuffled control. The paired ancestry-minus-depth advantage was 0.108 (95% cell-bootstrap interval, 0.077–0.135), positive in 45 of 47 cells. After removing broadcast, residual-energy capture was 0.527 for ancestry and 0.392 for depth bins. Degree–depth surrogates captured 0.578 of total energy; the smaller ancestry advantage over this control was 0.041 (0.020–0.062). Thus coarse geometry accounts for much of the capacity, with additional information in the retained ancestry.

The original eight cells showed the same ordering after common-mode correction. The Pinky second-mouse analysis also favored ancestry, but only eight of ten qualifying cells had enough input-bearing sites and candidate routes for the eight-column comparison. Those cells had just 9–13 input-bearing sites, and ancestry nearly saturated their residual fields. We retain their lower-budget comparisons and exclusions separately rather than pool them with the larger arbors. The earlier route-generated positive controls remain in Supplementary Figs. S10, S20 and S27; they demonstrate capacity under imposed alignment and are distinct from this independent-field extension.

Anatomical labeling and cost measures constrain the interpretation. Direct E/I labels covered 4.71% of inputs in the disjoint cohort; the 73.0% direct-versus-proxy agreement was measured among jointly labeled contacts in the original cohort. Nonzero site-by-route coefficients quantify a delivery-cost proxy, not feedback synapses or

metabolic expenditure. Geometry, labeling and compression sensitivities are reported in Methods and Supplementary Figs. S12, S25 and S33. The principal result is that ancestry provides spatial capacity beyond a shared broadcast, under matched common-mode access, with its size depending on the arbor and comparison dictionary.

Focal shunting sets gains on ancestry-defined credit routes

Morphology supplies spatial routes; conductance can regulate their amplitudes. We examined this connection by comparing a focal inhibitory shunt with a current injection matching its baseline focal current in reconstructed reciprocal cables. A separately solved somatic current restored baseline somatic voltage after either intervention. The endpoint was the change in the modeled excitatory-conductance gradient.

Let G be the passive conductance matrix and $\mathbf{b}$ the current-source vector, so that baseline voltages satisfy $G\mathbf{V} = \mathbf{b}$. With $\mathbf{e}_k$ the unit vector at compartment k , adding inhibitory conductance κ_k at compartment k changes both G and its source: $G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top$, $\mathbf{b}' = \mathbf{b} + \kappa_k E_I \mathbf{e}_k$. The matched injection supplies baseline current $\kappa_k(E_I - V_k)$ without changing G . For a fixed somatic loss derivative, the adjoint changes by

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (8)$$

The column $G^{-1} \mathbf{e}_k$ gives the voltage response to unit current at k . On a passive tree with a somatic loss, the resulting adjoint change has an exact ancestry structure: all descendants of the shunt share the attenuation factor $[1 + \kappa_k (G^{-1})_{kk}]^{-1}$, while each sister block along the path to the soma shares its own factor (Fig. 6A; SI S4). These disjoint blocks form an ancestry partition. Weighting their indicators by the baseline soma-to-site transfer gives a dictionary on which the shunt acts through diagonal gains, providing a physical instance of the framework's gain factor Γ_u . The diagonal form applies in this partition basis; it need not hold for an arbitrary selected set of overlapping routes.

We compared descendants $D(k)$ with off-route sites $C(k)$ at similar depth. A separate kernel criterion tests whether the adjoint change is larger below the shunt (Methods). The analyzed gradient was sensitivity to open excitatory conductance, $\gamma_i = q_i(E_E - V_i)$. The local driving force $E_E - V_i$ can change differently within each block, so the complete synaptic gradient need not share a single block gain. With primes denoting post-intervention values and ε_γ a numerical floor defined in Methods, we measured absolute log change, $m_i = |\log[(|\gamma'_i| + \varepsilon_\gamma)/(|\gamma_i| + \varepsilon_\gamma)]|$, and localization $\Lambda_k = \text{median}_{D(k)} m_i - \text{median}_{C(k)} m_i$. Positive localization means a larger gradient change below the shunt.

In the permissive normalized model, shunts changed descendant gradients more strongly than depth-matched off-route gradients, with increasing localization at higher dose (Fig. 6B,C). Current injection changed descendant gradients roughly threefold less. Its off-route changes were zero by construction under the soma-restoring control; this is not a general claim that current injection cannot change a gradient. Holding the post-shunt voltage but replacing its adjoint with the baseline adjoint removed most localization, supporting a transport contribution beyond the voltage change alone (Fig. 6D).

Physical calibration sharply limited the effect. At axial resistivity $R_a = 150 \Omega \text{cm}$ and specific membrane resistance $R_m = 15,000 \Omega \text{cm}^2$, mean shunt-minus-current localization was small and opposite-signed in the initial eight cells and a disjoint 45-cell cohort from the same mouse: -0.0019 and $+0.0024$, respectively. Lower membrane resistance increased the contrast by more than an order of magnitude (Fig. 6E). Distributed background conductance produced the same qualitative recovery at fixed high membrane resistance (Supplementary Fig. S11), consistent with a dependence on electrotonic state [25, 39].

Adding weak voltage-dependent channel currents produced a dose response close to the passive response at $R_m = 1,000 \Omega \text{cm}^2$ with one baseline-leak unit of distributed background conductance (SI S9). This checks the local linearization around solved steady states. The baseline Jacobian included gate derivatives but remained fixed across doses, so the check does not establish robustness to strongly active dendrites or channel kinetics.

Together, the exact partition identity and the electrical calibration connect the morphology-defined dictionary to conductance-dependent route gains [26]. Their spatial selectivity depends on cable state. These imposed, state-matched perturbations measure modeled transport, rather than endogenous error production, observed plasticity or a learning advantage.

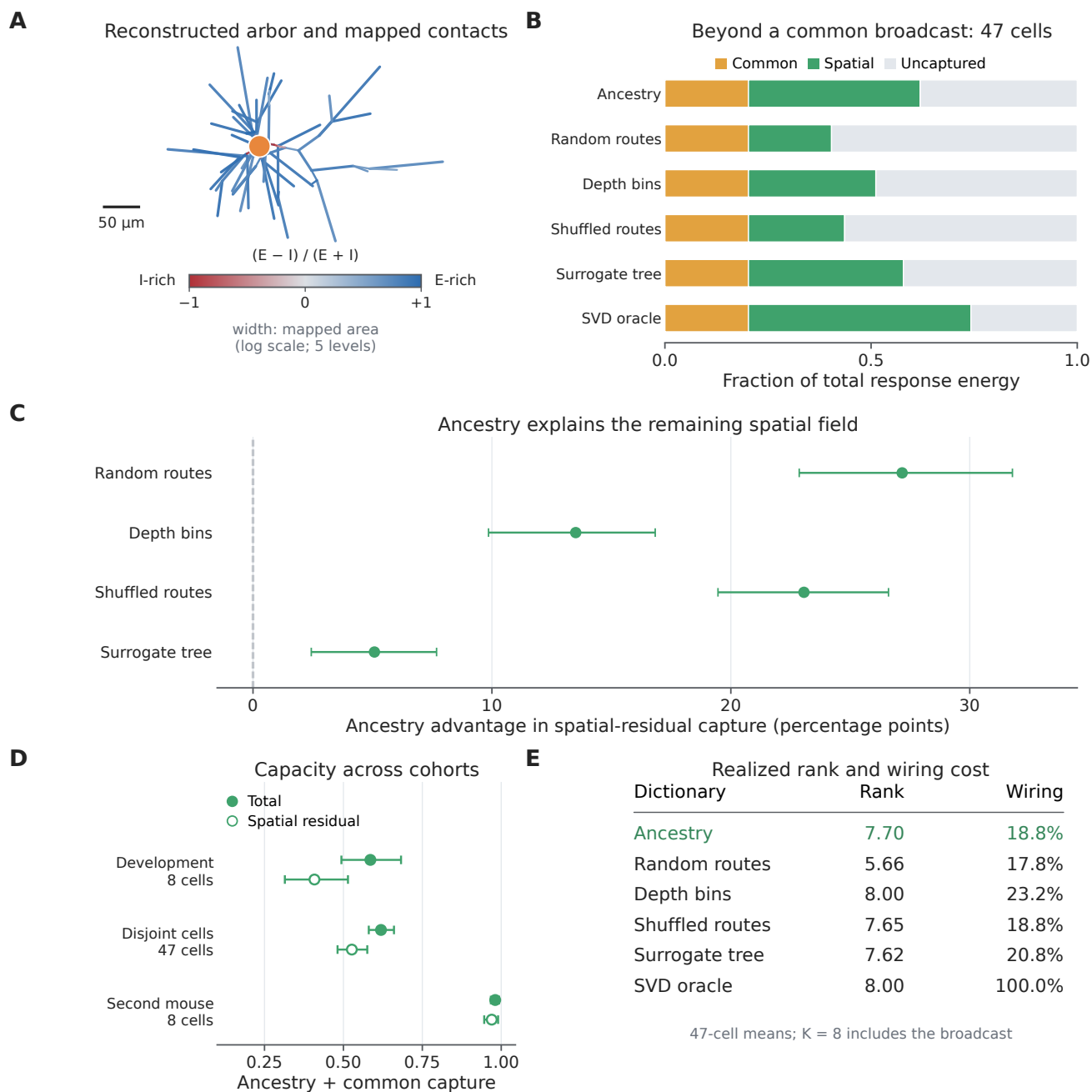


Figure 5: Ancestry captures spatial response structure beyond a shared broadcast. **A**, Median-sized reconstructed arbor from the initial cohort. Color indicates mapped E/I area balance; width indicates total contact area; scale bar, 50 μ m. **B**, Mean energy decomposition for modeled focal-shunt responses in 47 disjoint v661 cells from the same mouse. Every dictionary has one common broadcast and seven spatial columns. **C**, Paired ancestry advantages in residual energy capture after removing broadcast. Random routes sample anatomical sites; shuffling preserves column nonzero counts; surrogates preserve segment depths and parent out-degrees. **D**, Separate original, v661 and Pinky cohorts. Filled/open symbols indicate total/residual capture; Pinky's eight eligible cells contain only 9–13 excitatory-bearing sites each. **E**, Actual rank and nonzero density relative to dense eight-column wiring in v661. Equal column count does not ensure equal rank or wiring; ancestry and shuffling have identical nonzero counts. Randomized controls average 200 draws per cell. Symbols and bars show cell means and 95% bootstrap intervals; these describe within-cohort consistency. Capture uses squared excitatory-area-weighted field energy. The common-constrained SVD is an oracle capacity ceiling, and modeled fields are not observed teaching signals.

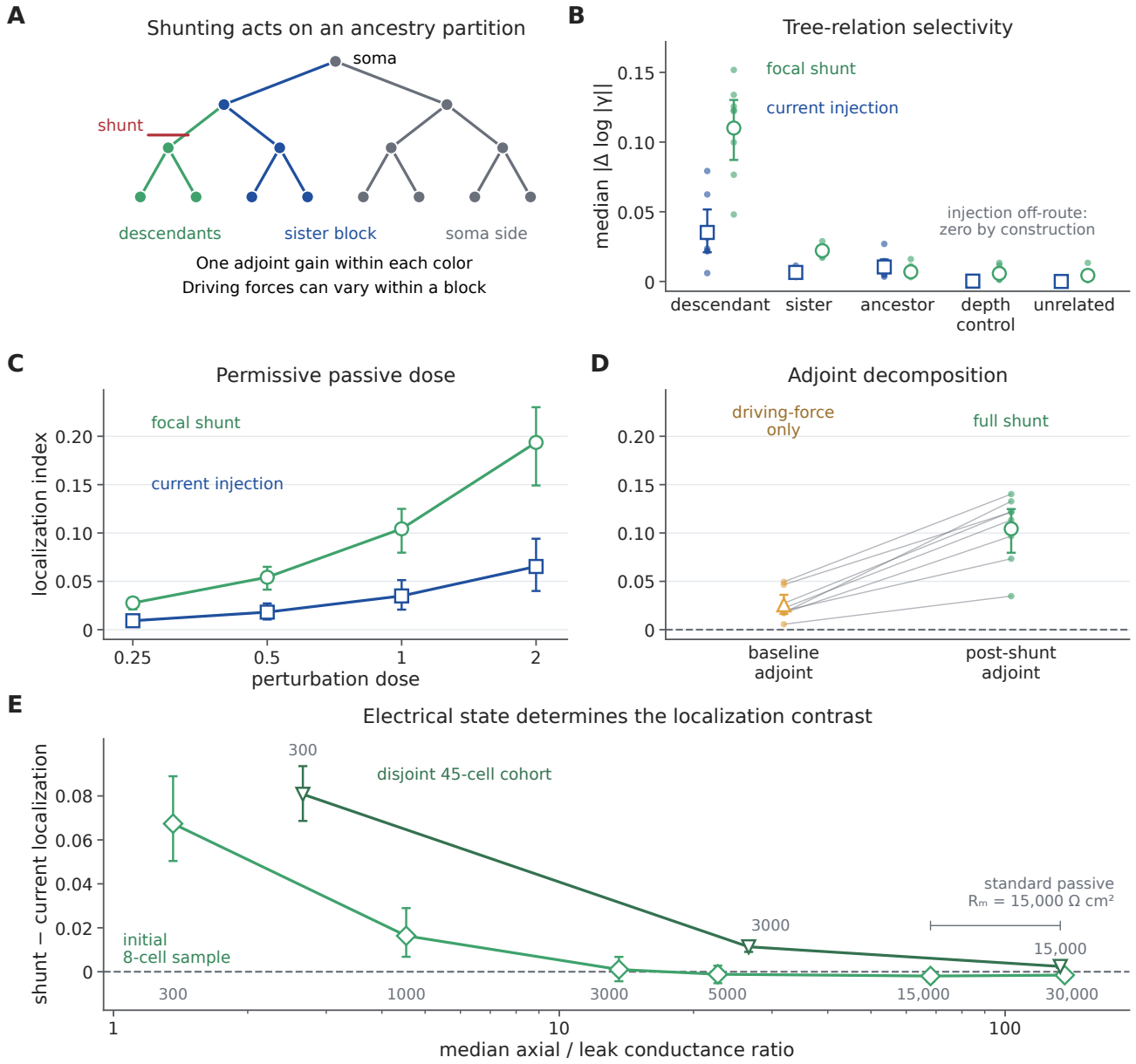


Figure 6: Focal shunts set ancestry-defined gains whose selectivity depends on electrical state. **A**, On a passive tree with a fixed somatic loss derivative, a shunt gives the same adjoint attenuation factor to every site in each colored ancestry block. The resulting gain is diagonal in the baseline-weighted partition dictionary; local synaptic driving forces can still vary within a block. **B**, Absolute log change in excitatory-conductance gradient magnitude by tree relation at unit dose, in eight reconstructed cells under permissive normalized passive parameters. A current injection matches baseline focal current, and a separately solved somatic current restores baseline somatic voltage after either intervention. Off-route injection zeros arise from this construction. **C**, Localization, defined as descendant minus depth-matched off-route median absolute log-gradient change, across doses relative to baseline local conductance. **D**, Keeping the post-shunt voltage while restoring the baseline adjoint removes most localization. Lines pair cells. **E**, Electrical calibration in the initial eight cells and a disjoint 45-cell cohort from the same mouse. The horizontal axis is the cohort-median axial/leak ratio at fixed axial resistivity $R_a = 150 \Omega \text{ cm}$; numerical labels give membrane resistance R_m in $\Omega \text{ cm}^2$. Shunt-minus-current contrasts are small and opposite-signed at $R_m = 15,000$, and increase at lower resistance. Small points are cells; symbols and whiskers are means and 95% cell-bootstrap intervals after averaging sites within cell. These state-matched perturbations measure modeled transport, not plasticity or learning benefit.

Measured responses test ancestry alignment and expose a transfer-geometry limit

The anatomical results identify candidate routes; measured responses ask whether functional relationships preferentially follow them. Seven MICrONS targets had complete arbors, mapped functionally imaged partners and repeated visual responses. The selected-scan dataset contained 69 connected imaged partners, and a scan-complete analysis included 13 eligible scans nested within the same seven targets. Recordings were obtained during stimulus presentation, not during learning.

The direct structure–function test asked whether partners with more shared dendritic ancestry had more similar condition-mean responses, after controlling Euclidean separation and soma-to-contact depth differences. Mean partial rank correlations were -0.069 for selected scans and -0.048 for all eligible scans; both target-bootstrap intervals included zero (Fig. 7A). This empirical test provides no support for preferential ancestry alignment in the sampled partners. Its scale differs from population-level like-to-like wiring [40].

A separate offline learning comparison used presynaptic responses to predict each target’s response through its complete passive tree. Trainable nonnegative excitatory conductances received input proxies at the partners’ dominant mapped contact segments; a linear somatic readout predicted the target. Stimulus identities were held out, and transformations and model choices used training data only. Exact learning supplied trial-specific spatial sensitivities. Each restricted rule instead projected the baseline soma-to-site transfer profile $\mathbf{v}_0$ into a four-route dictionary D , giving $\mathbf{v}_D = P_D \mathbf{v}_0$, and delivered

$$\hat{\delta}_{i,D}^V = \delta_i^{\text{out}} \mathbf{v}_D. \quad (9)$$

Here δ_i^{out} is the scalar output error on trial i . The readout and bias always received exact gradients; only input-conductance credit differed.

The realized dictionaries explain what this comparison can test. Selected subtree routes were predominantly distal supports: across scans, they reached 48.0% of input-bearing sites and contained an average of 1.31 sites per route. Six of thirteen scans had only single-site routes (Fig. 7B,C). Moreover, one unrestricted fixed baseline-transfer profile already reconstructed 98.4% of the exact field and 97.6% of exact updates. The model therefore offered little trial-dependent spatial variation for a richer encoder to exploit. Its learning comparison chiefly diagnoses loss of a nearly fixed transfer profile when that profile is restricted to incompletely covering supports.

Consistent with that geometry, the subtree rule had no reliable held-out advantage over random anatomical routes or site shuffles. Exact learning performed best, while ridge regression captured most of its predictability (Fig. 7D). Exact learning improved normalized MSE over ridge by 0.015 on average (95% target-bootstrap interval, 0.004–0.027). Here NMSE is relative to predicting the training mean, unlike the known population-variance normalization in the artificial tasks. Ten split/initialization replicates were averaged within scans and then within targets; the seven targets were the inferential units.

Allowing optimal trialwise coefficients within the same dictionaries barely improved ancestry update reconstruction (Fig. 7E). For trial-by-parameter update matrices U , we measured

$$\mathcal{M}_\Delta(D) = 1 - \frac{\|U_{\text{exact}} - U_D\|_F^2}{\|U_{\text{exact}}\|_F^2}, \quad (10)$$

where the Frobenius norm sums squared entries. Ancestry rose from 0.232 for its fixed profile to 0.244 with oracle coefficients, compared with 0.385 for random routes and 0.468 for shuffles. Thus the primary limitation was the selected span’s geometry, not coefficient freezing. The re-executed checkpoint discrepancies and exact oracle definitions remain in Methods and Supplementary Table S9.

Observed-input removal and noise controls quantify how response predictability depends on available partners (Supplementary Fig. S34). Separate imposed-alignment fields establish that the anatomical dictionaries can support learning when the target lies in their span (Supplementary Fig. S5). Neither supplies a positive control for endogenous teaching in the visual-response recordings. The measured ancestry-similarity test and the modeled transfer diagnostic together delimit what the present data can establish about biological use of the proposed routes.

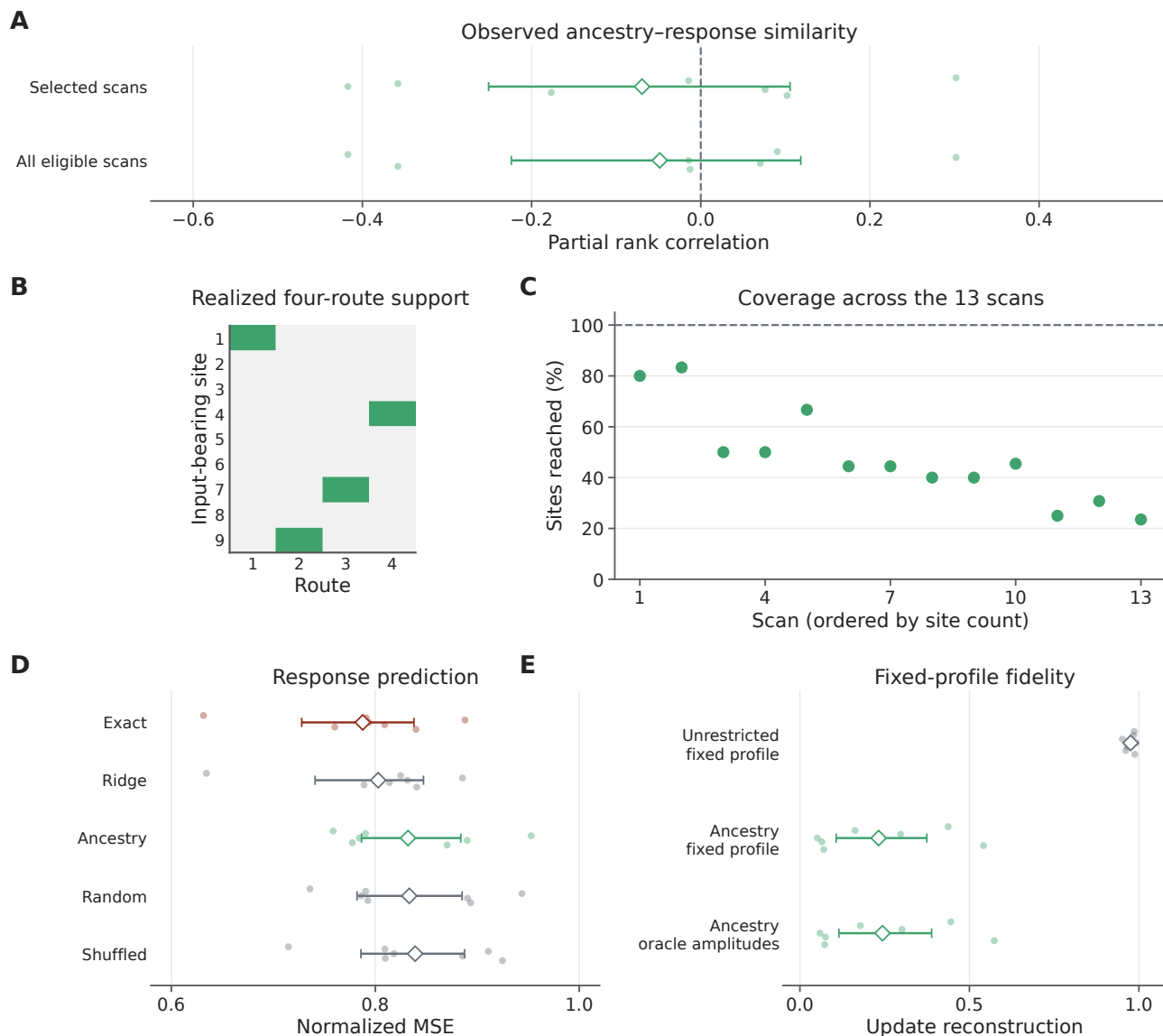


Figure 7: Observed ancestry alignment and a separate transfer-geometry diagnostic. **A**, Target-level partial rank correlations between shared ancestry and response similarity after controlling Euclidean separation and contact-depth differences. Selected scans contain 69 partners; all eligible scans comprise thirteen scans of the same seven targets. **B**, Actual site-by-route support in the scan with median input-bearing site count, chosen without response outcomes; green entries indicate nonzero routes. Site and route identifiers are retained in Source Data. **C**, Fraction of sites reached by selected ancestry routes in each scan, ordered by site count; scans are descriptive repeated observations within targets. **D**, Held-out normalized MSE for exact and restricted nonlinear learning and training-tuned ridge regression. NMSE uses the training-mean predictor as baseline. **E**, Common-checkpoint update reconstruction by the unrestricted fixed transfer profile, its restriction to ancestry routes, and optimal trialwise amplitudes within that ancestry span. A value of one is exact reconstruction. The last two differ in coefficient access, not in spatial support. In **A,D,E**, dots are seven target values; diamonds and whiskers show means and 95% target-bootstrap intervals. Panels **B–E** describe an offline model comparison and do not measure biological learning signals.

Discussion

Task structure changes which spatial distinctions a useful dendritic learning signal must preserve. The route-dictionary framework makes this question explicit: morphology supplies patterns of delivery, a teaching mechanism supplies their coefficients, and conductance controls their gains. The experiments form a progression from neuronal identity to branch selection, ancestry grouping and example-dependent credit across interacting branches. Their common lesson is that useful resolution is task- and learner-dependent; anatomical detail or exact-gradient fidelity alone does not determine the outcome.

Image tasks locate a substantial first step at the neuron boundary. Supplying distinct neuronal coordinates accounts for most of the performance gain, including when a random map provides between-neuron feedback. Contextual conflict then creates a need to distinguish active from inactive branches, a distinction that can be supplied either in feedback or by gating local eligibility. Hierarchical distractors create a smaller advantage for ancestry at limited bandwidth. Its four-route location follows from the generator's coefficients, making the experiment a test of a designed prediction about grouping rather than evidence for a universal optimum. These comparisons separate the information delivered from the physical substrate that carries it.

The new interaction comparison gives the strongest link between task structure and spatial credit. Pairwise and quartic targets with identical input-sensitivity spectra share an exactly compatible tree, yet produce different learned credit spectra and sharply different tolerance of fixed spatial profiles. Initial gain calibration and a common learning rate do not remove this difference. A nearly one-dimensional pairwise field can still have poor uniform-broadcast capture, so the relevant issue is the available spatial direction as well as its dimension. The measured spectra describe learned states; they cannot retrospectively validate the failed initialization selector. Nor does failure of the tested profiles establish that every compartment needs an independent external error. Intermediate dictionaries, adaptive profiles and alternative dynamics remain candidate solutions.

Forward structure sets the conditions under which a credit comparison is informative. The cut criterion identifies representationally compatible trees in the scalar multi-affine model, connecting input grouping to hierarchical tensor descriptions [41–43]. It also permits constructive structure estimation from noisy examples, retained here as supporting evidence. Conductance trees have their own grouping constraints, and a compatible positive-conductance teacher is learned well with simple credit. Serial composition likewise helps when distributed nuisance factors require successive cancellation, but its interaction with credit depends on optimization and training budget. These results explain why task rank, physical depth and feedback resolution should not be treated as interchangeable quantities.

Reconstructed arbors extend the dictionary proposal to measured geometry. Giving all controls the same common broadcast reveals ancestry capacity in the residual modeled field, including the disjoint 47-cell cohort. The smaller contrast against degree–depth surrogates identifies the contribution beyond coarse geometry. Focal shunting supplies a complementary link to biophysics: in a passive tree its adjoint effect reweights the shunt site's ancestry partition, while synaptic driving forces contribute additional local changes. The electrotonic boundary specifies when those changes remain spatially selective. This account concerns transport of sensitivity and differs from inhibition that computes errors or forward gain–load regulation [44–47].

The measured-response analyses delimit the biological claim. Ancestry–response similarity provides an empirical structure–function test and shows no preferential alignment in this small cohort. The offline learning comparison principally tests whether a restricted dictionary preserves an almost fixed cable-transfer profile; its sparse selected supports make it a weak test of endogenous route use. A direct biological experiment would compare plasticity below a focal inhibitory site with electrotonically matched off-route sites while matching somatic voltage, first-order current and the outcome signal. Varying conductance state would test the predicted separation between shared and ancestry-selective effects [22, 48, 49].

The framework can complement somatic prediction, segregated dendritic teaching, interneuron-mediated errors and burst-based plasticity [28, 30–32], as well as cable-learning and engineered gating models [35, 50–53]. Spikes, temporal eligibility, recurrence and full channel kinetics may change both the available profiles and how their coefficients are estimated [54, 55]. The present contribution is a way to ask, and experimentally separate, which distinctions the task demands, which routes the arbor offers, and which of them the learning dynamics can use.

484 Methods

485 The Methods define the models, task manipulations and inferential units behind the main comparisons. Supplementary
 486 Sections S1–S4 provide derivations and the interaction studies; S5 gives complete artificial-network protocols;
 487 S6–S12 cover anatomy, perturbations and measured responses. The supplementary cohort index lists all experiments.
 488 Resolved configurations, seeds, checkpoints, exclusions and source hashes accompany the numerical tables. Reusing
 489 a cohort for a diagnostic does not add independent observations.

490 Conductance models and delivery coordinates

491 In a directed tree, $[b_1, \dots, b_D]$ lists branching factors from soma to tips, giving $\sum_{d=1}^D \prod_{r=1}^d b_r$ nonsomatic compart-
 492 ments. A child c supplies output $a_c = f_c(V_c)$. With leak g_n^L , aggregate open excitatory/inhibitory conductances
 493 $G_n^X = \sum_{i \in \mathcal{S}_n^X} g_i^X x_i^X$ for contact class X , and child couplings $g_{c \rightarrow n}^{\text{den}}$, the steady voltage is

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + G_n^E E_E + G_n^I E_I + \sum_c g_{c \rightarrow n}^{\text{den}} a_c \right), \quad (R_n^{\text{tot}})^{-1} = g_n^L + G_n^E + G_n^I + \sum_c g_{c \rightarrow n}^{\text{den}}. \quad (11)$$

494 Here $x_i \geq 0$ is contact activity, $g_i \geq 0$ its conductance, and E_X its reversal potential. A terminal-to-soma pass
 495 evaluates the directed steady state. Artificial conductance trees use dimensionless leak one and reversals $E_L = E_I = 0$,
 496 $E_E = 1$. Monotone parameter transforms enforce nonnegative conductances. Branch-output nonlinearities, termed
 497 reactivations in the implementation, are identity functions when disabled. Both excitatory and inhibitory contacts
 498 learn in the regular-tree experiments.

499 For a reciprocal cable residual $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = 0$, the nonsingular voltage Jacobian J_V gives the adjoint

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}, \quad \partial \mathcal{L} / \partial g_i = -\mathbf{q}^T \partial \mathbf{F} / \partial g_i. \quad (12)$$

500 With passive current balance, the synaptic gradient is $x_i(E_i^{\text{rev}} - V_n)q_n$. This established sensitivity calculation supplies
 501 the reference for restricted credit [9, 10, 29, 35]. Directed LocalCA instead combines local eligibility with the
 502 supplied voltage-error field as in Eq. 3; supplying the exact path field recovers the exact selected gradients.

503 The raw-additive control retains the graph but sums signed inputs without the conductance denominator or
 504 reversal-potential driving force. Historical normalized-additive controls additionally standardize voltage across
 505 stage coordinates within each example; Supplementary Figs. S1–S4 identify those variants. The optional four- and
 506 five-factor rules add stage-level preconditioners and are described in Supplementary Section S1.

507 Writing a delivery dictionary as $A\Gamma$ separates spatial profiles in A from diagonal channel gains in Γ . Nonzero
 508 gains preserve the column span and oracle capture, while changing delivered coefficients and conditioning. A passive
 509 focal shunt admits this representation in its baseline-weighted ancestry partition; arbitrary overlapping dictionaries
 510 need not. Physical stage count, feedback rank, external error dimension and state-dependent path resolution are
 511 distinct resources (Supplementary Tables S13–S14).

512 Image feedback ladders and checkpoint diagnostics

513 For example i , the trained decoder supplies somatic errors $\delta_{i,0,u}^V$ across neurons u . The strict scalar is their mean unless
 514 its magnitude is at most 10^{-6} , in which case it is the largest-magnitude coordinate, preserving that coordinate’s sign.
 515 The same scalar reaches every arbor. Neuron-specific feedback repeats each neuron’s own coordinate throughout
 516 its arbor; exact transport multiplies it by the local path derivatives. The historical matched-width fallback retained
 517 neuronal identity at the 128-wide somatic stage, affecting 640 of 2,413,312 core parameters. A separately trained
 518 strict-scalar control tests that exception (Supplementary Table S16).

519 The main MNIST ladder uses 128-soma [3, 3] shunting and raw-additive networks with fifteen paired seeds per
 520 architecture. Within each architecture, contacts, decoder, initialization policy, optimization and validation-based
 521 checkpoint selection are shared across credit rules. Fashion-MNIST uses ten paired seeds; the raw-additive CIFAR-10
 522 ladder uses twenty and a 20-soma [3, 3, 3, 3] tree. Images are flattened without convolution or augmentation. The
 523 official test sets are held out; the 60,000-image training sets for MNIST and Fashion-MNIST are divided into 48,000

524 training and 12,000 validation examples. Exact schedules and architecture-specific initialization are retained in
 525 Supplementary Section S5 and resolved configurations.

526 Direct feedback alignment replaces the decoder derivative by a fixed Gaussian output-to-neuron matrix, scaled
 527 by inverse square root of output dimension and seeded by training seed. The factorial retains the same within-arbor
 528 rules and pairs new random-feedback fits with the frozen exact-readout reference. Ownership controls permute
 529 neuronal recipients without changing coordinate values or bandwidth.

530 Common-checkpoint diagnostics compare rules at identical weights and states. Exact path transport is checked
 531 against automatic differentiation. Gradient cosine, error magnitude by depth and mean-subtracted path energy
 532 are distinct summaries; none alone certifies a final learning outcome. The complete image diagnostic sheet is
 533 Supplementary Fig. S45. Historical depth and noise cohorts, including their task-identity and conductance-validity
 534 records, are specified in Supplementary Section S5.

535 Contextual conflict and ancestry routing

536 In the branch-conflict task, one context-selected branch determines a logistic output while all B branches form
 537 ungated local eligibilities. Nonselected Fashion-MNIST views are replaced independently by opposite-class images
 538 with probability χ . Exact feedback is $\delta_i \mathbf{1}[b = b_i^*]$; shared feedback is δ_i/B , where δ_i is output error and b_i^* the
 539 selected branch. Twenty paired seeds cross $B = 2, 4, 8$ with eight conflict doses. The algebraically equivalent
 540 gated-point calculation and BP receive the same updates; cyclic derangement preserves rank and support while
 541 changing the recipient.

542 Under centered, symmetric class evidence, shared-mode compatibility is $s(\chi) = 1 - 2\chi(B - 1)/B$, with zero at
 543 $\chi_c = B/[2(B - 1)]$. This mean-field prediction is separate from a trained accuracy crossing. The latter is reported both
 544 by interpolation of the mean curve and by each seed’s first sampled at-or-below-chance dose, retaining non-crossings.
 545 Checkpoint re-execution reproduces endpoints and compares exact/shared fields at both common and own trained
 546 states (Supplementary Fig. S29).

547 The ancestry factorial contains eight eight-dimensional blocks. Context c_i and binary label sign s_i are balanced,
 548 and inputs are $\mathbf{x}_{i,b} = a(b, c_i) s_i \mathbf{t}_b + \boldsymbol{\epsilon}_{i,b}$, with seed-specific unit teachers $\mathbf{t}_b$ and Gaussian noise $\boldsymbol{\epsilon}_{i,b} \sim \mathcal{N}(0, 1.1^2 I_8)$.
 549 The selected coefficient is one; sibling, same-half nonsibling and opposite-half coefficients are $-0.15, -0.45, -0.75$.
 550 A leaf permutation changes task-to-route alignment without changing the streams. For budget K , ancestry feedback
 551 has context-by-block entries $\Phi_{cb}^{(K)} = \sqrt{K/8} \mathbf{1}[g_K(\pi(c)) = g_K(b)]$, where g_K indexes contiguous groups and π maps
 552 streams to leaves. Thus every row has unit Euclidean norm.

553 The update is $\Delta \mathbf{w}_b = -(8\eta/N) \sum_i \delta_i \Phi_{c,b}^{(K)} \mathbf{x}_{i,\pi^{-1}(b)}$, with N training examples and rate η . Each seed supplies
 554 2,048 training, 2,048 test and 1,024 switch examples. Full-batch descent uses rate 0.3, eighty training epochs
 555 and twenty switch epochs. Controls include deranged ancestry, depth-interleaved and random-sparse supports,
 556 dense rank- K fields and a training-informed PCA reference. The best matched control is the per-seed post-training
 557 maximum over the four non-PCA controls. Integer test counts preserve ties in the named Wilcoxon comparison;
 558 Holm correction covers four budgets. Alternative mean-sign-flip and historical BH analyses remain explicitly labeled
 559 in Supplementary Table S2. The separate noisy-context encoder and cue-delay experiments are fully specified in
 560 Supplementary Section S5.

561 Update utility and prospective selection

562 Let the exact gradient be $\boldsymbol{\mu}$ and the delivered update estimator have mean $\mathbf{m}$ and covariance Σ_{route} at the current
 563 weights. For an L -smooth loss and a nonnegative step η , expected loss after the update is bounded above by $\mathcal{L} -$
 564 $\eta \boldsymbol{\mu}^T \mathbf{m} + (L\eta^2/2)(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})$. Optimizing this expression gives utility $U = [\boldsymbol{\mu}^T \mathbf{m}]_+^2 / [2L(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})]$,
 565 with zero for a zero denominator. Utility certifies one-step progress under the stated moments; it is not a long-horizon
 566 ordering.

567 Synthetic quadratic screens isolate signal retention, noise rejection and fixed gains over sixteen Haar coordinates,
 568 without a forward neuron (Supplementary Fig. S46). Their matched signal/noise hierarchy is a designed crossover.
 569 The trained factorial’s retrospective utility uses actual full-batch updates and a global logistic-curvature bound;

choosing maximum bandwidth matches its selection success. The separate prospective linear experiment retains all 12,800 fits, the initialization selector’s loss to fixed maximum-budget and rank-only baselines in every seed, and follow-ups forecasting evolving weights and noise. The 25,600-fit dynamics extension and observed-context-count control are reported in Supplementary Figs. S35 and S38–S39, with sealed choices, complete methods and computational costs.

Interaction compatibility, fixed profiles and learned spectra

For a subtree leaf set S , C_S arranges the target’s Fourier interaction coefficients by products inside and outside S ; C_S° removes the constant inside row. Scalar multi-affine trees represent the target exactly if and only if every centered cut has rank at most one. If $\mathcal{S}(T)$ contains all proper cuts of tree T and singular values are ordered from largest to smallest,

$$\text{MSE}(f, \hat{f}_T) \geq B(T; f) := \max_{S \in \mathcal{S}(T)} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (13)$$

Overlapping cut tails cannot be added as a lower bound. Supplementary Section S4 gives the constructive proof, minimum-depth algorithm and its relation to hierarchical tensor representations. The twelve-tree census, noisy-query estimator, end-to-end learning pipeline, conductance grouping controls and Boolean truth tables remain in Supplementary Figs. S36–S44. The theorem’s assumptions do not extend automatically to general conductance ancestors.

The matched credit experiment uses twenty fresh seed blocks after three development seeds select rates with equal tuning budgets. Matching and quartic conditions share tree, leaf assignment, initialization, inputs, additive noise and minibatch order. Exact, unit, initially calibrated and sign-only profiles use the same local eligibilities; calibration averages initial path derivatives over 256 unlabeled examples. Selected and common-rate outcomes are retained for SGD and Adam, giving 432 development and 1,200 fresh fits. Algebraic coefficients remain within $[-2, 2]$, and all bound contacts are recorded. Fits stop at 1,024 steps, with seven saved checkpoints. Noisy held-out NMSE divides by clean target variance: one for matching and nested, one-half for quartic. All have noise variance 0.0225. The primary Adam contrast subtracts the pairwise calibrated-minus-exact gap from its quartic counterpart.

For evaluation-example-by-site path matrix Q , the best fixed profile is the leading right singular vector. Its capture is $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$, an oracle reconstruction with example-dependent amplitudes. Additional diagnostics evaluate uniform, calibrated and ancestry profiles, loss-residual weighting, eligibility-weighted updates and root-mean-square normalization of each site. Participation-ratio effective rank is $(\text{tr} C)^2 / \text{tr}(C^2)$ for uncentered $C = Q^\top Q / N$. At fixed weights and inputs, changing labels leaves the path field unchanged; eighty relabeling controls verify this. Independent checkpoint analysis and faithful replay of 600 earlier fits provide numerical and trajectory checks. Complete definitions, pairing audits, optimizer effects and the positive-conductance boundary are in Supplementary Section S4.

Serial depth, sensor alignment and optimization

Physical-depth networks contain 64 somata and eight nonsomatic modules each. D1–D4 reallocate these modules across branching vectors [8], [2, 3], [2, 1, 2] and [1, 1, 2, 2]. Contacts and resources are matched; serial composition changes. Grouped-point controls connect every module directly to the soma. Raw-additive controls retain the graph without division, while flexible MLP controls are approximately parameter matched.

Excitatory blocks carry class evidence multiplied by lognormal nuisance factors. Inhibitory blocks sense those factors with alignment α : zero removes trial-specific information while preserving the sensor distribution, and one reports the factor. The generator floors noisy input rates before multiplying them by gains. Nested, flat and local-ratio families distinguish distributed hierarchical support from generic factor cancellation and locally available ratios. Full generators, floors, sensor-placement controls and contact inventories are defined in Supplementary Section S5.

Ten paired seeds support each original cohort. The initial maximum is 180 epochs with patience thirty; validation selects model states. BP and LocalCA optimizer recipes differ in learning rates and parameter-group treatment, so their endpoint gaps are reported separately. Exact-path LocalCA computes the exact selected gradients. An audit

of one hundred original trajectories verifies every configuration and final metric against its source: all fifty D3 fits across the five relevant credit/optimizer arms reached the epoch cap with validation loss still falling. The original depth contrasts are therefore fixed-budget outcomes. The longer-budget replay protocol and its checkpoint records are retained separately from the original cohort; it restarts original seeds because historical files contain no optimizer state.

Anatomical dictionaries and common-mode controls

Observed MICrONS skeletons and mapped contacts define soma-rooted branch-to-branch segments. Initial eight and disjoint 47-cell cohorts come from the same mouse; Pinky is a second mouse, with ten of twelve selected cells meeting the inherited typed-contact criterion. Each subtree route contains the input-bearing descendants of its origin. Contact labels are direct where available and otherwise proxy-based as specified in Supplementary Sections S6–S8. Compression preserves path length but only approximates resistance; direct-label, mapping and geometry sensitivities are retained.

The independent field consists of reciprocal passive-cable gradient changes induced by small focal shunts. Every corrected dictionary receives one constant column and $K - 1$ spatial columns. Ancestry selection uses inhibitory fraction times excitatory-area-weighted descendant fraction without field outcomes. Depth bins are reparameterized with the same constant while preserving their span; random routes, independent site shuffles and degree–depth surrogates provide controls. The weighted common component is removed before computing residual-energy capture. A common-constrained SVD oracle supplies the rank-budget ceiling. Numerical rank, coverage and nonzero density are reported separately; equal column count alone does not match those resources.

The original eight cells are post hoc development for common-mode correction. The protocol was frozen before evaluating the new independent-generator outcomes in v661 and Pinky. Two hundred randomized controls are averaged within each cell. At eight columns, all 47 v661 cells and eight Pinky cells qualify; other eligible budgets are retained without substituting for missing eight-column outcomes. Paired cell-bootstrap intervals and Holm-adjusted comparisons describe within-cohort consistency, not animal-population replication. The older route-generated positive controls use separate fields and endpoint definitions (Supplementary Section S7).

Focal shunting and the electrotonic boundary

Up to sixteen focal sites per cell span depth quartiles and require at least three excitatory descendants and three comparison sites. Normalized E/I scales are 0.35 with reversals one and -0.2 . Shunt doses are 0.25, 0.5, one and two times local membrane-plus-synaptic conductance. A matched injection supplies first-order current $\kappa_k(E_I - V_k)$ without changing the conductance matrix. A separately solved somatic current restores baseline voltage and is held fixed during differentiation. The squared-loss target is one normalized unit below baseline; finite differences check the adjoint gradients.

Localization subtracts depth-matched off-route from descendant medians of $|\log[(|\gamma'_i| + \epsilon)/(|\gamma_i| + \epsilon)]|$, with $\epsilon = 10^{-15} \max(1, \max_i |\gamma_i|)$. Because this is unsigned, signed gradient changes are reported separately. Adjoint and driving-force substitutions in $\gamma_i = q_i(E_E - V_i)$ distinguish transport from local voltage effects. Sites and relation-reassignment templates are nested within cells. The passive ancestry-partition identity and its dictionary qualification are derived in Supplementary Section S3.

Physical calibration uses cylindrical axial conductance $\pi r^2/(R_a \ell)$ and leak $2\pi r \ell/R_m$, expressed in nS, with spherical soma area $4\pi r^2$. Here r and ℓ are segment radius and length, R_a axial resistivity and R_m specific membrane resistance. Sweeps include $R_a = 100, 150, 250 \Omega \text{cm}$ and $R_m = 300\text{--}30,000 \Omega \text{cm}^2$, fixed and locally normalized doses, and background conductance. The disjoint calibration cohort contains 235 eligible sites in 45 cells. Geometric fallbacks and dose definitions are specified in Supplementary Section S9.

The weak-channel check uses $R_a = 150$, $R_m = 1,000$ in those units and background conductance equal to leak. Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times excitatory conductance. Sixty-four draws per cell passed the predefined steady-state checks. Responses use rank-one updates of the fixed baseline Jacobian, including voltage-gate derivatives, followed by somatic state matching. They

are local linearizations rather than newly solved nonlinear equilibria at each dose (Supplementary Fig. S47).

Measured-response prediction and geometric diagnostics

The seven-target cohort includes thirteen eligible scans. Repeated responses are averaged by stimulus condition for the ancestry-similarity analysis; partial rank correlations control Euclidean separation and soma-to-contact depth differences. Offline learning holds out stimulus identities. Training-only percentile transforms produce nonnegative presynaptic proxies, and each partner enters at its dominant mapped contact segment. A linear somatic readout predicts the standardized target response. Only input-conductance credit differs across rules; readout and bias gradients remain exact.

Four-route dictionaries are selected without target responses and receive the projected baseline transfer profile, as in Eq. 9. Realized supports are reconstructed directly from saved segment identifiers. The displayed matrix uses the scan with median site count, with identifier-based tie breaking; all scan supports are retained. At common exact-trained states, field and eligibility-weighted update reconstruction compare the fixed profile, per-trial field projection and the best within-span update. These are oracle diagnostics rather than newly trained rules. The 130 exact re-executions retain their endpoint discrepancies.

Ridge regularization is selected within training data for every outer split, using the same measured predictors. Observed-input removal, predictor noise and reliability masks are separate sensitivity analyses with no held-out information used for selection. Ten split/initialization replicates are averaged within scans and then within seven targets. Imposed-alignment quadratic fields and the six-animal external signed-credit reanalysis are separate datasets and tests, fully described in Supplementary Sections S10–S12.

Statistics and computational provenance

Confirmatory choices precede outcome inspection but are not described as public preregistration. Pilots, post hoc analyses and same-seed reruns are identified separately. Effect sizes accompany experiment-specific 95% intervals, or adjusted 97.5% intervals where prespecified. Tests are two sided unless a directional protocol states otherwise. Named multiplicity families and exact tie treatment are retained in Source Data; null results imply equivalence only when a margin is tested.

Independent units are paired training seeds, reconstructed cells, postsynaptic targets or external animals. Sites, contacts, scans, splits, checkpoints and randomized draws remain nested. Seed counts reflect the experiment’s purpose; no prospective parametric power calculation was performed. Source hashes certify retained inputs and implementations, not unavailable historical runtime states. Supplementary Section S13 lists all exclusions, statistical families, numerical checks and release records.

OpenAI Codex assisted with code, statistical and consistency audits, figure preparation, literature checks and language editing. The authors remain responsible for design, interpretation, numerical accuracy, citations and the final manuscript.

Ethics and data reuse

No new experiments involving animals or human participants were performed. All biological analyses reused public MICrONS, DANDI, and published animal source data. The source studies describe the original data collection and institutional approvals [22, 38, 56].

Data availability

MICrONS anatomy and connectivity are available through the public CAVE datastack minnie65_public. The primary cohort used materialization 1822, skeleton-service reconstructions, and the synapses_pni_2 table. The disjoint sensitivity cohort used the public v661 static release. Secondary target-structure labels used only in the full-contact analysis came from the in-development synapse_target_predictions_ssa_v2 table. These labels

are distributed only as derived data and remain subject to the source-data access terms. All principal structural findings were repeated without them.

The independent-animal replication used the MICrONS phase-1 layer-2/3 Pinky v185 fixed meshes, soma-valence table, and dense synapse table archived at Zenodo DOI 10.5281/zenodo.3710459 [57]. File sizes, Zenodo MD5 checksums, and local SHA-256 checksums are recorded in the supplied manifest. Volumetric imagery is archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as DANDI:000402. The supplied manifest records the exact asset and table identifiers, and the source publication describes the full data release [38].

Public benchmark datasets are available from their original repositories. The inhibitory-census tables and SWCs are available with the source record for Schneider-Mizell et al. [56]. The Francioni et al. workbook is available through the Source Data link associated with the Nature article [22]; its checksum and the sheet names used here are recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and non-restricted summaries are supplied with this manuscript. Authentication tokens and externally hosted raw caches are not redistributed.

Code availability

The accompanying reviewer software archive contains the complete training implementation, exact-gradient and perturbation analyses, versioned configurations, tests, figure scripts, and byte-identified snapshots of the critical training components under the MIT License. A separate Source Data archive contains the numerical data underlying every display item. These archives are available to editors and reviewers with the submission. The same version will be released publicly under the MIT License in a versioned repository and permanent archival record upon publication.

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Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the theory of local credit assignment, its implementation, and its interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.